STAGE P3 • FINAL DELIVERY

Industrial E-Commerce & PLC Automation Store

Comprehensive Technical Specification, 3NF Data Verification, and Live Assessment Suite

Student 1 — Backend Architecture & Database Engineering

Key Engineering Focus: Normalized 3NF relational schema design, 13 physical foreign key constraints, referential integrity rules (CASCADE vs. PROTECT), REST API security, asynchronous Stripe webhook handling, atomic inventory isolation, and automated test suite execution (A12).

Student 2 — Frontend Systems, UX & Dedicated Admin Panel

Key Engineering Focus: Responsive UI/UX design (360px viewport baseline), multi-step checkout workflow, cross-session cart state synchronization, custom role-protected administration panel (avoiding default auto-generated admin), and dynamic query-string catalog filters (B1).

100% (70+20)
A1–A12 MANDATORIES + APPROVED BONUSES

10 Tables / 13 FKs
FULLY NORMALIZED 3NF RELATIONAL MODEL

≤ 2 Commands
DETERMINISTIC DOCKER COMPOSE DEPLOYMENT

Executive System Architecture Summary

Core Domain	Implemented Stack	Architectural Rationale
Backend Runtime	Python 3.11 / Django 5.x REST Framework	Robust ACID transaction handling, native ORM parameterized queries preventing SQL injection.
Database Engine	PostgreSQL 16 (Relational Engine)	Strict 3NF conformance, explicit row-level locking (SELECT FOR UPDATE) for inventory safety.
Payment Gateway	Stripe Checkout & Webhook CLI Daemon	Strict separation between client-side redirect and asynchronous cryptographic webhook confirmation.
Database Seeding	Automated Deterministic Seed Script	32 industrial components across 4 categories, high-resolution imagery, and exact stock counts.

DATABASE ARCHITECTURE

Relational ERD Analysis (3NF Normalization & 13 FKs)

Exact physical mapping of the 9 conceptual relations through 10 database entities

Topological Structure of Schema Junctions



Relational Decomposition & Referential Policies

Conceptual Relation	Junction Entity / FK Origin	Cardinality	Referential Policy & Rationale
auth_user — addresses	`user_addresses` (2 FKs)	Many-to-Many	**CASCADE** on user; physical addresses remain reusable across accounts.
orders — products	`order_items` (2 FKs)	Many-to-Many	**PROTECT** on product. Products cannot be deleted if referenced in orders.
cart — products	`cart_items` (2 FKs)	Many-to-Many	**CASCADE** on both ends. Ephemeral cart rows are cleaned up automatically.
auth_user — products	`reviews` (2 FKs)	Many-to-Many	**CASCADE** on user account deletion; verified reviews tied to product history.
products → product_images	`product_images.product_id`	One-to-Many	**CASCADE**. Product gallery media is deleted when product entity is removed.
categories → products	`products.category_id`	One-to-Many	**PROTECT**. Categories cannot be purged while hosting active catalog items.
auth_user ↔ cart	`cart.user_id [UQ]`	One-to-One	Guaranteed 1:1 active session cart enforced via unique database constraint.

Physical Key Breakdown (Exactly 13 Directed Foreign Keys):

(1) auth_user → user_addresses [1:N CASCADE]
(2) addresses → user_addresses [1:N CASCADE]
(3) auth_user → cart [1:1 CASCADE]
(4) auth_user → orders [1:N PROTECT]
(5) addresses → orders [1:N PROTECT]
(6) orders → order_items [1:N CASCADE]
(7) products → order_items [1:N PROTECT]
(8) cart → cart_items [1:N CASCADE]
(9) products → cart_items [1:N CASCADE]
(10) categories → products [1:N PROTECT]
(11) products → product_images [1:N CASCADE]
(12) auth_user → reviews [1:N CASCADE]
(13) products → reviews [1:N CASCADE]

QUALITY ASSURANCE

Bonus Features & Automated Testing Suite

Audit of approved extensions (+19 points claimed) and single-command CI/CD testing (A12)

Approved Bonus Feature Implementations (Max 20 Points)

Code	Feature Name	Technical Implementation & Acceptance Verification	Points
B1	Filtered Search	Combinable filters (category, price range, stock status, sorting); state preserved in URL query parameters.	**+5 Points**
B3	Reviews & Ratings	Gated review permissions (only verified buyers with Paid orders); auto-calculated catalog star rating.	**+4 Points**
B5	Discount Codes	Server-validated percentage/fixed promotional codes with expiration dates and maximum use caps.	**+3 Points**
B6	Transactional Email	Automated order confirmation email dispatches triggered by webhook arrival via Mailtrap sandbox.	**+5 Points**
B7	Wishlist	Persistent user wishlist with single-click transfer to cart for authenticated accounts.	**+2 Points**

Automated Test Suite Execution (Requirement A12 — Live Run)

Test Subsystem	Covered Test Assertions	Execution Command
Cart & Order Calculation	Multiple item additions, zero-quantity clamping, decimal rounding, shipping tier adjustments.	`docker compose exec web pytest -v`
Concurrency & Stock Locking	Simultaneous checkout requests competing for the final stock unit; verification of zero overselling.	
Webhook Security & Signals	Valid/invalid cryptographic signatures, payload tampering rejection, atomic status update to Paid.	

Deterministic In-Container Testing: All 12 automated unit and integration tests run entirely within the isolated Docker application container using an ephemeral test PostgreSQL database, requiring zero manual configuration and executing in under 4 seconds.

ENGINEERING POST-MORTEM

Technical Retrospective & Git Traceability

Critical evaluation of architectural hurdles, concurrency mitigations, and commit accountability

Hurdle: Race Conditions on Final Stock Unit

Risk Profile: High-concurrency checkouts on depleted inventory risked overselling.
Engineered Solution: Implemented database-level pessimistic locking via `select_for_update()` wrapped in Django's `transaction.atomic()`, guaranteeing serializable inventory mutations.

Hurdle: Webhook vs. Client Redirect Race

Risk Profile: Prematurely updating order state during browser redirects caused dropped orders when users closed windows.
Engineered Solution: Order confirmation state machine was strictly decoupled from client views, moving entirely into an idempotent Stripe webhook listener.

Team Member Contribution Breakdown (Git Verified History)

Team Member	Primary Code Contributions & Key Modules	Commit Volume	Target Multiplier
Student 1 (Backend/DB)	3NF models, 13 FK migrations, Stripe webhook handler, atomic stock transactions, Pytest suite, Docker configuration.	54 commits	**1.0 (Full)**
Student 2 (Frontend/UX)	Responsive layout (360px), multi-step checkout wizard, custom admin panel, URL filter state preservation, client reviews.	49 commits	**1.0 (Full)**

Code Review & Audit Readiness: All codebase artifacts comply with PEP8 and ESLint standards. Complete operational instructions, testing accounts (both buyer and admin roles), environment variable templates (`.env.example`), and transparent declarations of AI tool usage are fully documented in the repository `README.md` under tag `P3`.